

Visual Search Strategy and Information Processing Mode: An Eye-Tracking Study on Web Pages under Information Overload

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Abstract. Most studies about visual search on Web page focused on factors such as information form or information layout. However, there are few about information quantity. This study investigates the search strategy and information processing mode of visual search on Web pages under information overload. Results show that 1) users' search strategies on Web pages with and without information overload are the same which is paying more attention on picture. Information overload make this picture-oriented strategy more significant. 2) Under information overload, users are more likely to use "deep search" information processing mode which is decreasing parallel processing and increasing serial processing. These results indicate that (1) information overload doesn't change but impact users' search strategy; (2) Information overload has a significant impact on users' information processing mode. This study provides some evidence both for cognitive psychology and human-computer interaction, especially Web page design.

Keywords: Information overload, Visual search, Web pages, Search strategy, Information processing mode.

1 Introduction

Internet has become one of the most important parts in people's daily life. As the carrier of on-line information, Web page is the main interface of human-computer interaction (HCI) on Internet. Researchers of HCI have described an adaptive and intelligent HCI that could predict and diagnose users' cognitive state, and then adapt itself to improve HCI [1]. To achieve this, it is necessary to investigate users' physical

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and mental activities. Traditional methods of user behavior study on Web pages (e.g. feedback analyzing, data mining and user modeling) are “off-line” investigation. It’s impossible to obtain users’ cognitive activities through these methods. Eye-tracking technology allows researchers to record users’ eye movement data when they interact with Web pages. These data including fixation, eye movement contrail and pupil size could directly and objectively reflect users’ behavior on Web pages, which could reveal their cognitive activities [2].

There are two types of visual behavior on Web pages: visual search and browsing. Visual search refers to the behavior of searching on Web pages with a target, while browsing refers to the behavior of browsing on Web pages without target [3,4]. Previous studies have demonstrated that users’ visual search on Web pages is impacted by many factors, such as colors [5,6], fonts [7], information forms [8,9] and information layout [10]-[16]. However, there are few studies focused on the impact of information quantity. Nowadays, users often face on Web pages with massive information. To deal with it, Web page designers have to know how users react when they search on Web pages under information overload.

In this study, an experiment with two groups was designed using eye-tracking to investigate users’ search strategy and information processing mode of visual search on Web pages under information overload.

2 Experiments

2.1 Experimental Design

Ten different Chinese Web pages involving different topics (e.g. cell phone or clothing) were used in this study. We controlled information quantity by limiting the size of Web pages. In Group 1, Web pages were single-page without scrollbar which could be presented on one screen; in Group 2, Web pages were double-page with vertical scrollbar, half of which could be presented on one screen. As two groups used same information density (e.g. font, font size and line space), the information quantity in Group 2 was approximately twice as much as that in Group 1.

To eliminate the effects of information form and location, we used two target forms (text and picture) and arranged them on different locations. There were 10 different targets including 5 texts and 5 pictures. Texts were 3 to 6 Chinese characters which were companies’ name (e.g. Dell) and pictures were companies’ logo (e.g. the logo of Audi). Targets in Group 1 were arranged on 5 locations: upper-left, lower-left, central, upper-right and lower-right; Targets in Group 2 were arranged on 10 locations: the 5 locations on upper half of Web page and that on lower half. Thus, there were 10 Web pages in Group 1 (5 locations $\times$ 2 target forms) and 20 in Group 2.

2.2 Participants

There were 130 undergraduates or postgraduates from various majors in this study, half of which were female. All participants were native Chinese speakers, right-handed, skilled users of Internet and had normal or corrected-to-normal vision. 50 participants with age range of 21 to 25 ($M = 23.0$, $SD = 1.3$) were in Group 1 and 80 with age range of 20 to 27 ($M = 23.2$, $SD = 1.3$) were in Group 2.

2.3 Apparatus and Procedure

Eye movements were recorded at the rate of 120 HZ by Tobii T120 eye-tracker, which had a 17 LCD monitor with resolution set to 1024 × 768 pixels. Web pages were automatically presented on screen by software called Tobii Studio.

At the beginning of each task, the pre-page which had description of the target was presented (whether target was text or picture, it was described in Chinese so that participants didn't know the target form). Participants needed to click into the search page, found the target and clicked on it. After that, participants could have a rest and click into the next task.

3 Results

We firstly contrast the search time of two groups. The search time is defined as the time between Web page showing up and participant clicking on the search target. As shown in Fig. 1, the search time of Group 2 is significantly longer than Group 1, which means that the search efficiency of Group 2 is significantly lower than Group 1. These result indicates that the increase of information quantity reduces users' visual search efficiency, which is consistent with Hornof's finding [17,18] and Rau's [19]. Thus, we consider Group 1 as non information overload and Group 2 as information overload.

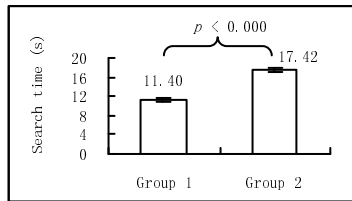


Fig. 1. The search time of two groups

3.1 Search Strategy

To investigate participants' search strategy under information overload, we look into their eye movement contrails. Fig. 2 shows some typical contrails (because some contrail has too many fixation points, it only shows no more than 20 fixation points) with different search target forms and locations under information overload.

It is likely that participants tend to pay more attention on picture than text under information. To find out what it likes under non information overload and how information overload impacts, we further count the distribution of the earliest 20 fixation points under non information overload and information overload, as shown in Fig. 3.

No matter information overload or not, the number of fixation point on picture is significantly larger than text. However, the number of fixation point on picture under information overload is significantly larger than non information overload. These results indicate that information overload doesn't change participants' search strategy which is paying more attention on picture but makes this picture-oriented search strategy more significant.

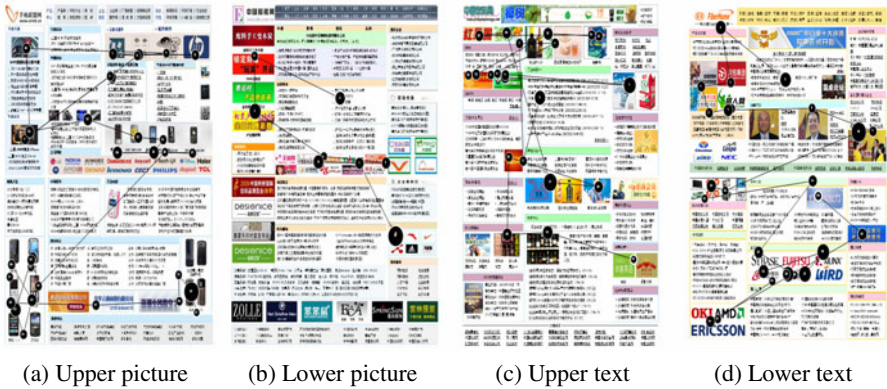


Fig. 2. Some typical eye movement contrails with different search target forms and locations. (a) Target is picture on upper half of Web page. (b) Target is picture on lower half of Web page. (c) Target is text on upper half of Web page. (d) Target is text on lower half of Web page.

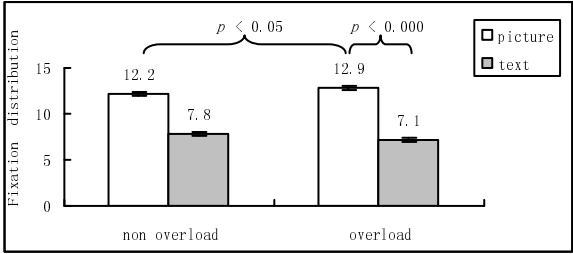


Fig. 3. The distribution of the earliest 20 fixation points

3.2 Information Processing Mode

There are two basic information processing type for visual search: parallel processing and serial processing [20]. Parallel processing refers to the behavior that comparing search target and all the distracters at the same time. Serial processing refers to the behavior that comparing search target and distracters one to another. When user searches target on a Web page, he uses parallel processing to find a target area in which he thinks the target is, and then uses serial processing to search in this area. If he doesn't find the target, he uses parallel processing again to find another target area and then uses serial processing again [21]. The average fixation duration could show how much participants use serial processing: the more the average fixation duration is, the more participants use serial processing. The fixation count could show how much participants use parallel processing: the more the fixation count is, the more participants use parallel processing.

To investigate the impact of information overload on information processing mode, we analyze participants' the average fixation duration and fixation count. The average fixation duration is defined as "fixation duration / fixation count", which refers the average duration of each fixation points. Fixation count is the number of fixation points.

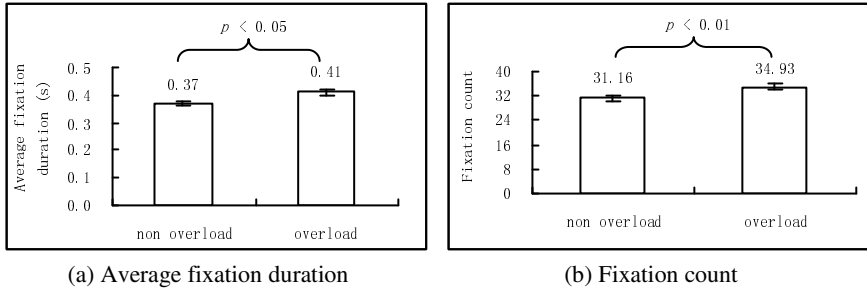


Fig. 4. The average fixation duration and fixation count of two groups. (a) Average fixation duration. (b) Fixation count.

As shown in Fig. 4a, the average fixation duration under information overload is significantly longer than non information overload, which means that participants increase serial processing under information overload. Although the fixation count under information overload is significantly more than non information overload, the increasing rate (12.1%) doesn't match that of information quantity (approximately 100%). The rate of fixation point / information quantity under non information overload is 31.16 per page; however, the rate under information overload is about 17.47 per page. So, the number of fixation points decreases under information overload, which means that participants decrease parallel processing.

4 Discussion

The studies about information forms have reported that picture has the superiority effect to text [8]. Carroll et al. through eye-tracking experiment found participants look at picture information first then read text information [22]. Rayner et al. in their study about print advertisements also reported that participants paid more attention on picture than text [23]. In our study, no matter participants faced on Web pages under information overload or not, their search strategy was paying more attention on picture, which is consistent with previous studies. This picture-oriented search strategy become more significant under information overload, which suggests that important information should be presented as picture among plenty of information.

Based on common knowledge we assume that when participants search on Web pages under information overload, they would increase their saccade speed (jump from one fixation point to another faster) and decrease the duration of fixations in order to improve their search efficiency. This "fast saccade" information processing mode would increase parallel processing and decrease serial processing. However, results show that participants are more likely to decrease parallel processing and increase serial processing. With this "deep search" information processing mode, participants search an area more carefully, move to another and rarely come back. They use this mode which could be explained as an effect of "inhibition of return" to prevent searching area has been searched in order to improve their search efficiency. This information processing mode accords with cognitive economy principle.

5 Conclusion

This study investigates the search strategy and information processing mode of visual search on Web pages under information overload using eye-tracking. Results show that information overload doesn't change users' search strategy which is paying more attention on picture, but it makes this picture-oriented strategy more significant. Under information overload, users are more likely to decrease parallel processing and increase serial processing.

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